

Write Equations

Lesson 7-1

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$x + 5 = 12$ means 'some number plus 5 equals 12'

Equation

A math sentence with an equal sign showing both sides are the same.

In $n + 3 = 10$, the letter n stands for the unknown number ($n = 7$)

Variable

A letter that stands for an unknown number.

$7 + 3 = 10$ — both sides equal 10

Equal sign

The = sign, showing both sides are the same.

$2x + 5$ is an expression; it becomes an equation when you write $2x + 5 = 15$

Expression

A math phrase with numbers and letters, but no equal sign.

Key Ideas & Notes

- Detective Ruiz received a tip: 'A number of stolen gems plus 8 more equals 20 total gems.' She needs to write an equation to represent this clue before she can solve the case.
- How would you write this situation as an equation?
- Use the balance scale to see how each word problem translates into an equation. Drag the correct expression to each side of the scale to make it balance.

Think About It

- What quantity is unknown in this situation?
- What does 'plus 8 more' tell you about the operation?
- What does 'equals 20 total' tell you?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Detective Ruiz's clue says 'a number of stolen gems plus 8 more equals 20 total.' How would you turn this clue into an equation, and what does each part stand for?

Level 1 support

Start here: I let a variable stand for the unknown number of gems.

The unknown is ____, **so I write the variable** ____.

La incógnita es ____, así que escribo la variable ____.

My equation is ____ **because 'plus 8' means** ____ **and 'equals 20' means** ____.

Mi ecuación es ____ porque 'más 8' significa ____ y 'es igual a 20' significa ____.

Word bank: equation variable equal sign expression

Listen for: Students choose a variable (such as n) for the unknown gems, build $n + 8 = 20$, and connect 'plus 8 more' to $+ 8$ and 'equals 20 total' to $= 20$.

Level 2

Why does the equation $n + 8 = 20$ represent the clue better than the expression $n + 8$ alone? Justify using the role of the equal sign.

- The equation is better than the expression because ____.
- The equal sign tells me ____ that an expression cannot show.

EXPLORE

On the balance scale you put $n + 8$ on one side and 20 on the other. How did you decide what belongs on each side of the equal sign?

Level 1 support

Start here: Both sides of the equation must represent the same total.

I put ____ on the left side because ____.

Puse ____ en el lado izquierdo porque ____.

The equal sign means both sides are ____.

El signo igual significa que ambos lados son ____.

Word bank: equation variable equal sign expression

Listen for: Students place the expression $n + 8$ on one side and the total 20 on the other, explaining the equal sign means both sides have the same value.

Level 2

Compare 'twice a number equals 20' with 'a number plus 8 equals 20.' How would the equations differ, and why?

- 'Twice a number' would be ____ instead of ____ because ____.
- The operation changes the equation because ____.

CONNECT

When would writing an equation help you solve a real problem instead of just guessing?

Level 1 support

Start here: An equation organizes what I know and what I want to find.

Writing an equation helps me because ____.

Escribir una ecuación me ayuda porque ____.

I would write an equation to find ____.

Escribiría una ecuación para hallar ____.

Word bank: **equation** **variable** **expression** **equal sign**

Listen for: Students explain that an equation captures the relationship clearly, lets you use a variable for the unknown, and can be solved instead of guessing.

Level 2

Critique: 'Any sentence with numbers is an equation.' Use the words expression and equal sign to explain why this is wrong.

- This is wrong because an expression ____.
- A sentence is only an equation when it has ____.

Guided Examples

Example 1

Which equation represents: 'A number plus 12 equals 30'?

Solution: 'Plus' means addition, so the equation is $n + 12 = 30$.

Answer: A. $n + 12 = 30$

Example 2

Which equation represents: 'Seven less than a number is 18'?

Solution: 'Seven less than a number' means start with the number and subtract 7: $n - 7 = 18$.

Answer: A. $n - 7 = 18$

Example 3

Which equation represents: 'Three times a number equals 21'?

Solution: 'Three times' means multiply by 3, so the equation is $3n = 21$.

Answer: A. $3n = 21$

Write About the Math

The Writing Revolution

I can explain my equation using the words equation, variable, equal sign, and expression.

1. Kernel Sentence subject + verb

Model: Equation is a math sentence with an equal sign showing both sides are the same.
Ecuación es una oración matemática con un signo igual que muestra que ambos lados son iguales.

Write a kernel sentence about equation. Use a subject and a verb.

Escribe una oración base sobre ecuación. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Equation matters in math
Ecuación importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Equation matters in math because ____.
Ecuación importa en matemáticas porque ____.

but
pero

Equation matters in math, but ____.
Ecuación importa en matemáticas, pero ____.

so
entonces

Equation matters in math, so ____.
Ecuación importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about equation.
Di un hecho verdadero sobre equation.

Equation ____.

Question
Pregunta

Ask a question about equation.
Haz una pregunta sobre equation.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about equation.
Muestra entusiasmo sobre equation.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with equation.
Dile a un compañero qué hacer con equation.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The unknown is ____, **so I write the variable** ____.

La incógnita es ____, *así que escribo la variable* ____.

My equation is ____ **because 'plus 8' means** ____ **and 'equals 20' means** ____.

Mi ecuación es ____ *porque 'más 8' significa* ____ *y 'es igual a 20' significa* ____.

I put ____ **on the left side because** ____.

Puse ____ *en el lado izquierdo porque* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which equation represents: 'A number divided by 4 equals 8'?

A. $n / 4 = 8$

B. $4n = 8$

C. $n - 4 = 8$

D. $n + 4 = 8$

Show your work:

2. A detective found some fingerprints on Monday and 14 more on Tuesday, for a total of 30. Which equation represents this?

A. $f + 14 = 30$

B. $f - 14 = 30$

C. $14f = 30$

D. $f / 14 = 30$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write a real-world situation that can be modeled by the equation $n / 5 = 9$. Explain what n represents and verify your equation makes sense.

Sentence starter: Situation: _____. In this problem, n represents _____. Check: $n = \underline{\hspace{1cm}}$ because $\underline{\hspace{1cm}} \div 5 = 9$.

Show your work:

Reflect — Exit Ticket

Which equation represents: 'A number divided by 6 equals 9'?

- A. $n / 6 = 9$
- B. $6n = 9$
- C. $n - 6 = 9$
- D. $n + 6 = 9$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** $A. n / 4 = 8$ — *'Divided by 4' means division: $n / 4 = 8$.*
2. **Try It 2:** $A. f + 14 = 30$ — *The detective's Monday fingerprints (f) plus 14 more equals 30: $f + 14 = 30$.*
3. **Exit Ticket:** $A. n / 6 = 9$ — *'Divided by 6' means division: $n / 6 = 9$. Check: $54 / 6 = 9$ ✓*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equation is a math sentence with an equal sign showing both sides are the same.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).